

Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

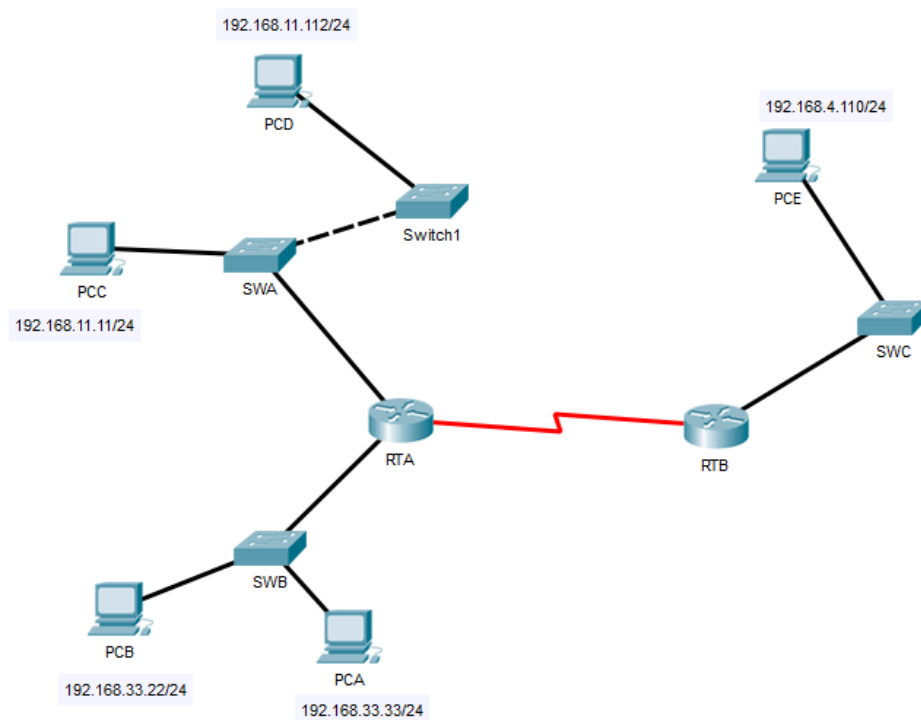


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

Figure 1.1 Router RTA

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1       -          0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

Figure 1.2 Router RTB

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0002.4a00.0e91    DYNAMIC Fa0/1
1       000c.8546.7d85    DYNAMIC Fa1/1
SWA#
```

Figure 1.3 Switch SWA

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a    DYNAMIC Fa0/1
SWB#
```

Figure 1.4 Switch SWB

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614    DYNAMIC Fa0/1
SWC#
```

Figure 1.5 Switch SWC

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

Figure 1.6 PCA

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

Figure 1.7 PCB

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>|
```

Figure 1.8 PCC

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

Figure 1.9 PCD

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

Figure 1.10 PCE

- e. What are your thoughts on the results?

No entry for all PCs. It is because all the PCs did not receive any ARP request. There will be no entry in the PC ARP table.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=12ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

- In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746        dynamic
```

- In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

```
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.33         0002.1755.9a06        dynamic
```

- In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

```
C:\>arp -a
No ARP Entries Found
```

Figure 2.1 PCC

```
C:\>arp -a
No ARP Entries Found
```

Figure 2.2 PCD

```
C:\>arp -a
No ARP Entries Found
```

Figure 2.3 PCE

Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time=16ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 5ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.11.1          0002.4a00.0e91       dynamic
```

- Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
    Internet Address      Physical Address      Type
    192.168.11.1          0002.4a00.0e91       dynamic
    192.168.11.112        0001.6462.0278       dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=11ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 4ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91        dynamic
192.168.11.112        0001.6462.0278        dynamic
```

- e. Discuss the results you got from all the commands on PCC.

The ARP entries are now can be seen in the ARP table only after PCC pings PCB, PCD, and PCE. No ARP entries are found before pinging all of these PCs. The ARP table displays entries once PCB, PCD and PCE receive the ARP request from PCC and subsequently receive pings from PCB, PCD AND PCE.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91 ARPA    FastEthernet1/0
Internet  192.168.11.11     3          00D0.D39A.C0D9 ARPA    FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A ARPA    FastEthernet0/0
Internet  192.168.33.22     3          0060.47EA.A746 ARPA    FastEthernet0/0
RTA#
```

Figure 2.4 Router RTA

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 1 0060.702D.7C08 ARPA FastEthernet0/0
RTB#
```

Figure 2.5 Router RTB

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       0001.6462.0278   DYNAMIC   Fa1/1
1       0002.4a00.0e91   DYNAMIC   Fa0/1
1       000c.8546.7d85   DYNAMIC   Fa1/1
1       00d0.d39a.c0d9   DYNAMIC   Fa2/1
SWA#
```

Figure 2.6 Switch SWA

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
```



```
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       000c.cf0c.593a   DYNAMIC     Fa0/1
"
```

Figure 2.7 Switch SWB

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0001.977a.b614   DYNAMIC     Fa0/1
1       0060.702d.7c08   DYNAMIC     Fa1/1
SWC#
```

Figure 2.8 Switch SWC

- d. Do switches use an arp table? (Y/N)

Yes

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches maintain the ARP table that maps IP addresses for devices on the network. This table is used to determine the destination MAC address required for communication with other devices in the network.

- f. What information does the command ***show mac-address-table*** gives?

The term “DYNAMIC” refers to MAC addresses that the switch has automatically learned during its normal operation, as seen in the MAC address table.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a central switch S1 connected to three PCs (PC1, PC2, PC3) and a wireless router WRS2. PC1 is connected to S1 F0/11 (VLAN 10), PC2 to S1 F0/18 (VLAN 20), and WRS2 to S1 G0/1. WRS2 is connected to PC3. The addressing table on the right shows the IP configuration for each device.

Subinterfaces

- G0/0.10 172.17.10.1/24
- G0/0.20 172.17.20.1/24
- G0/0.88 172.17.88.1/24

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

PT Activity: 00:12:52

Time Elapsed: 00:12:52

Completion: 100/100

1/1

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

PT Activity: 00:42:48

Step 1: Connect the Internet interface of WRS2 to S1.
Connect the WRS2 Internet interface to the S1 F0/7 interface.

Step 2: Configure the Internet connection type.

- Click WRS2 > GUI tab.
- Set the Internet Connection type to Static IP.
- Configure the IP addressing according to the Addressing Table.

Step 3: Configure the network setup.

- Scroll down to Network Setup. For the Router IP option, set the IP address to 172.17.40.1 and the subnet mask to 255.255.255.0.
- Enable the DHCP server.
- Scroll to the bottom of the page and click Save Settings.

Step 4: Configure wireless access and security.

- At the top of the window, click Wireless. Set the Network Mode to Wireless-N Only and change the SSID to WRS_LAN.
- Disable SSID Broadcast and click Save Settings.
- Click the Wireless Security option.
- Change the Security Mode from Disabled to WPA2 Personal.
- Configure cisco123 as the passphrase.
- Scroll to the bottom of the page and click Save Settings.

Part 2: Configure a Wireless Client

Step 1: Configure PC3 for wireless connectivity.
Because SSID broadcast is disabled, you must manually configure PC3 with the correct SSID and passphrase to establish a connection with the router.

- Click PC3 > Desktop > PC Wireless.
- Click the Profiles tab.
- Click New.
- Name the new profile Wireless Access.
- On the next screen, click Advanced Setup. Then manually enter the SSID of WRS_LAN on Wireless Network Name. Click Next.
- Choose Obtain network settings automatically (DHCP) as the network settings, and then click Next.
- On Wireless Security, choose WPA2-Personal as the method of encryption and click Next.
- Enter the passphrase cisco123 and click Next.
- Click Save and then click Connect to Network.

Step 2: Verify PC3 wireless connectivity and IP addressing configuration.

- The Signal Strength and Link Quality indicators should show that you have a strong signal.
- Click More Information to see details of the connection including IP addressing information.
- Close the PC Wireless configuration window.

Time Elapsed: 00:42:48 Completion: 100/100

☐ Top ☐ Dock 1/1

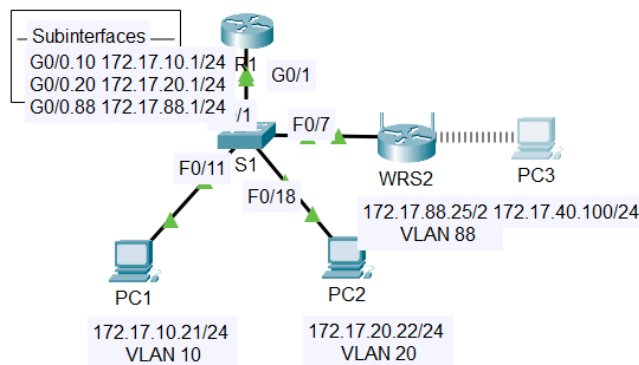
Cisco Packet Tracer - C:\Users\safiy\Downloads\Lab 4 SECR1213-20250116\lab4ASafiya.pkt - Guest - 20...

File Edit Options View Tools Extensions Window Help

Logical Physical x 444 y 361 Root 08:51:30

Time: 00:55:45 Realtime Simulation

Copper Straight-Through



Step 1: Connect the Internet interface of WRS2 to S1.
Connect the WRS2 Internet interface to the S1 F0/7 interface.

Step 2: Configure the Internet connection type.

- Click WRS2 > GUI tab.
- Set the Internet Connection type to Static IP.
- Configure the IP addressing according to the Addressing Table.

Step 3: Configure the network setup.

- Scroll down to Network Setup. For the Router IP option, set the IP address to 172.17.40.1 and the subnet mask to 255.255.255.0.
- Enable the DHCP server.
- Scroll to the bottom of the page and click Save Settings.

Step 4: Configure wireless access and security.

- At the top of the window, click Wireless. Set the Network Mode to Wireless-N Only and change the SSID to WRS_LAN.
- Disable SSID Broadcast and click Save Settings.
- Click the Wireless Security option.
- Change the Security Mode from Disabled to WPA2 Personal.

Time Elapsed: 02:37:51 Completion: 100/100

☐ Top ☐ Dock 1/1